DTFT Frequency Assignment for Selective Hodge–Kuramoto Lock

with Full Use of the Harmonic Space and Normal-Mode Frequencies

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Abstract

In `paper_v2` every oscillator on a harmonic bin has natural frequency $\omega_i = 0$ (Eq. 14b); we prove this forces full lock for any connected coupling, which is what the real 2B4T+Qwen run reported ($\langle r \rangle = 1$, 147/147 locked). Following the author's normal-mode framework, the natural frequencies are *defined* by the generalized eigenproblem $(K - \omega^2 M)v = 0$ with stiffness the weighted Hodge 1-Laplacian and mass the Hodge star $\star_1$, and *measured* per model by the DTFT of its band coefficients in MRH order (off-grid peak, Candan/Jacobsen estimator, Goertzel/chirp-z); the lattice dispersion relation $\omega = 2c\sin(\pi\nu)$ ties the two. Offsets enter the `jev_kuramoto` tick through R-KTE-7; with the KTE linear lift the pair lock condition is exactly $|\Delta\omega| \le 2KW_{\text{lock}} = 1/128$ turn/tick (≈ 2 cycles of drift across channels). We use the harmonic space fully: $\ker L_1 \cong H^1$ with Betti number derived for the MRH-digit torus, gauge-invariant period fingerprints, integer winding tests, elastic P/S scaling of the exact and coexact parts, Forman–Bochner curvature corrections, and a heap-key persistence filtration that orders the anneal. Constants are derived from the author's files or the literature; the rest are listed as tuning. Only arithmetic sanity checks are reported.

1 Problem: the $\omega = 0$ degeneracy

paper_v2 Eq. (15) ticks $\theta_i \leftarrow \theta_i + \sum_j K_{ij}\sin(\theta_j - \theta_i)$ with $\omega_i = 0$.

Proposition 1 (uniform $\omega \Rightarrow$ full lock)**.** *Let $K_{ij} = K_{ji} \ge 0$ define a connected graph and $\omega_i \equiv \omega$. In the co-rotating frame $\dot\theta = -\nabla V$, $V = -\frac{1}{2}\sum_{ij} K_{ij}\cos(\theta_j - \theta_i)$; the synchronized manifold is the global minimum, asymptotically stable, and every initial condition inside an open half circle converges to it [1]. For the KTE linear lift (R-KTE-1/8) the tick is $\theta \leftarrow (I - L_K)\theta$ on lifted differences, and with $\lambda_{\max}(L_K) < 2$ (paper_v2 Gershgorin bound) every non-constant mode contracts by $|1 - \lambda_k| < 1$, so $\theta \to c\mathbf{1}$, $r \to 1$, and every node meets the R-KTE-2 window.*

Sketch. V is analytic and bounded, so the gradient flow converges (Łojasiewicz). In a half circle all $\cos(\theta_j - \theta_i) > 0$, so the Jacobian $-L_{K\cos}$ is a connected weighted Laplacian with simple kernel $\mathbf{1}$. The linear-lift case is the stated affine map. □

So $r = 1$ with n/n locked carries no information; the log line "$r = 1.000000$, locked 147/147, merged 49/49" is the predicted outcome, not agreement between models.

2 The harmonic space, used fully

2.1 Complex and Betti number

paper_v2 orders each channel axis $N = \prod_j r_j$ by mixed-radix digits and treats each digit level as circulant (Eq. 4): the index grid is the product of cycles $T = \prod_j C_{r_j}$ (C_2 a single edge). With

d_0, d_1 the coboundaries and $L_1 = d_0 d_0^\top + d_1^\top d_1$ [7, 8, 9]:

$$\mathcal{H} = \ker L_1 \cong H^1(T;\mathbb{R}), \qquad \beta_1 = |E| - \operatorname{rank} d_0 - \operatorname{rank} d_1. \tag{H1}$$

Proposition 2 (the 2-cells decide)**.** *(a) No 2-cells: $\beta_1 = |E| - |V| + 1$ — nearly every cochain is "harmonic". (b) All grid squares filled: $\beta_1 = D_{\geq 3}$, the number of digit levels with $r_j \geq 3$ (Künneth, $H^1(C_r) = \mathbb{R}$ iff $r \geq 3$). For 2560×6912 ($5 \cdot 2^9$, $3^3 \cdot 2^8$): $\beta_1 = 4$. (c) The complete bipartite channel graph $K_{m,n}$ of paper_v2 Sec. 1: $\beta_1 = (m-1)(n-1)$ without 2-cells, 0 with all 4-cycles filled.*

Sanity (exact rational rank, `dtft_lock_sanity.py`): $C_5 \times C_3$ gives $\beta_1 = 2$ with squares and 16 without. Hence the "low bins" of paper_v2 Eq. (5) are a *quasi-harmonic band*, not $\ker L_1$; the true classes (b) carry the topological invariants (Sec. 2.3), the band carries the oscillators.

2.2 Basis, spectral gap, projections

With unit or per-level stars on the square-filled torus, L_1 acts on each direction component as the scalar Laplacian, diagonalized by the DFT with $\lambda_k = \sum_j (2 - 2\cos\frac{2\pi k_j}{r_j})$ (paper_v2 Eq. 4). True harmonics are the $k = 0$ components; the band is

$$\mathcal{H}_\epsilon = \{(a,k) : \lambda_k < \lambda_{\text{gap}}\}, \qquad \lambda_{\text{gap}} = \min_{j: r_j \geq 3} \big(2 - 2\cos\tfrac{2\pi}{r_j}\big), \tag{H2}$$

the spectral gap of the coarsest long cycle: modes smoother than every non-trivial single-digit oscillation. This turns paper_v2's "lowest bin of each digit" into a derived rule. Projections: $P_{\mathcal{H}_\epsilon} = \mathcal{F}^{-1} \mathbf{1}_{\mathcal{H}_\epsilon} \mathcal{F}$, $P_{\text{grad}} = d_0 L_0^+ d_0^\top$, $P_{\text{curl}} = d_1^\top L_2^+ d_1$ [7, 10].

2.3 Fingerprints: periods, holonomy, class matching

With one loop γ_a per long digit, the period vector of a 1-cochain x_i is

$$p_{i,a} = \oint_{\gamma_a} x_i. \tag{H3}$$

Lemma 1. *$\oint_\gamma d_0 \alpha = 0$ on closed γ, and coexact parts integrate to zero on harmonic representatives of the cycles, so p_i depends only on the cohomology class of x_i: gradient changes (reorder/scale) and coexact changes (merge conflict) leave it fixed.*

Digit-cyclic shifts are automorphisms of T and leave p invariant (phase correlation still recovers them); arbitrary permutations are not, so they change the class (prediction P2). For phases, the winding number around γ_a,

$$q_a(\theta) = \tfrac{1}{2\pi} \textstyle\sum_{(u \to v) \in \gamma_a} \operatorname{wrap}(\theta_v - \theta_u) \in \mathbb{Z}, \tag{H4}$$

is invariant under continuous evolution while every edge difference stays below π (twisted states of ring Kuramoto [5]; higher-order Kuramoto on simplicial complexes [6]). A merge requires $q(\theta_i) = q(\theta_{\text{base}})$.

2.4 Gradient and curl spectra; data-derived stars

The split $E_g : E_c : E_h$ of each delta is a diagnostic for the protocol. With stars, $L_1^\star = d_0 \star_0^{-1} d_0^\top \star_1 + \star_1^{-1} d_1^\top \star_2 d_1$ [7, 8]. *DFT constraint:* $L_1^\star$ commutes with every digit shift iff the stars are constant on shift orbits, i.e. depend only on digit level and edge direction (separable, circulant per level). Derived per-level stars: $\star_1^{(j)} = \overline{|d_0^{(j)} M_{\text{base}}|^2}$ (energy-normalized L^2 metric [11]), $\star_0 = \star_2 = 1$. Fully inhomogeneous stars (activation norms) break diagonalization; then the DFT with per-level averaged stars is used as a preconditioner for LOBPCG [22] on the band only.

2.5 Resizing and persistence

Whitney prolongation and de Rham restriction are cochain maps with $\mathcal{R}\mathcal{P} = I$ [11, 12], so they are isomorphisms on H^1 between widths keeping the long digits: periods and windings transport unchanged, and dropping a level with $r_j \geq 3$ removes exactly one class (record it in BNZ4). Since fixed 2-cell choices are degenerate (Prop. 2), squares enter a filtration at the max 12-bit MRH heap key of their edges (paper_v2 Eq. 7): low-error structure closes first. The persistent Laplacian's nullity is the persistent Betti number [15, 16], computed by standard persistence [13, 14]. The anneal (paper_v2 Eq. 18) couples, at stage s, only classes alive at s, longest-lived first; classes with persistence below one key LSB (1/4096, the key quantization) are noise and never coupled. The filtration ends at regime (b), so the β_1 true classes anchor the final stage.

3 Natural frequencies from normal modes (the author's tensor-calculus framework)

Small oscillations. The author's framework treats the band coefficients as coordinates q^j on a configuration manifold with mass tensor M_{ij} and stiffness K_{ij}: $M_{ij}\ddot{q}^j + K_{ij}q^j = 0$, so $q = v\,e^{i\omega t}$ with

$$(K - \omega^2 M)v = 0, \qquad K = \star_1 d_0 \star_0^{-1} d_0^\top \star_1 + d_1^\top \star_2 d_1, \quad M = \star_1 \tag{N1}$$

(Goldstein [18], Ch. 6). Multiplying $L_1^\star v = \lambda v$ by $\star_1$ gives exactly (N1) with $\omega_k = \sqrt{\lambda_k}$; K is symmetric PSD and M SPD, so the ω_k are real and the modes M-orthogonal. **Consequence.** True harmonic modes have $\omega = 0$ for *every* metric: the degeneracy of paper_v2 is intrinsic to $\ker L_1$. Oscillators that can discriminate must therefore live on the quasi-harmonic band, where $\omega_k > 0$ and depends on the model through its metric. **Defined vs measured.** On a circulant level with star ratio $c_j^2 = \star$-stiffness/$\star$-mass, $\lambda = 4c_j^2 \sin^2(\pi\nu)$, so

$$\omega(\nu) = 2c_j \sin(\pi\nu) \qquad \text{(lattice dispersion)}. \tag{N2}$$

The DTFT (Sec. 4) measures a model's dominant spatial frequency $\hat{\nu}_i$ off-grid; (N2) converts it to the normal-mode frequency the eigenproblem defines. Per model, first-order perturbation of (N1) with that model's stars gives the frequency shift $\delta(\omega_k^2) = v_k^\top(\delta K - \omega_k^2 \delta M)v_k$ (Rayleigh); the two estimates must agree to within the estimator variance, which is a consistency check in the protocol. **Elastic P/S split.** The Navier–Cauchy equation $\rho\ddot{u} = (\lambda + 2\mu)\nabla(\nabla\cdot u) - \mu\nabla\times\nabla\times u$ (isotropic Hooke, Lamé λ, μ [19]) separates, with $u = \nabla\phi + \nabla\times\psi$, into P-waves $c_P^2 = (\lambda + 2\mu)/\rho$ (divergence, exact part) and S-waves $c_S^2 = \mu/\rho$ (curl, coexact part); fields with neither are topological. The discrete analogue is $K = c_P^2 K_{\text{down}} + c_S^2 K_{\text{up}}$, justifying separate frequency scalings for the gradient and curl components and none for harmonics. Only the ratio c_P/c_S affects the lock (overall scale is absorbed by tick scaling L1); we derive it per tensor as $\sqrt{R_g/R_c}$, the ratio of the base's energy-weighted mean Rayleigh quotients on the exact and coexact parts. **Curvature.** The covariant wave operator $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ on a 1-form obeys the Weitzenböck identity $\Delta_{\text{Hodge}} = \nabla^*\nabla + \text{Ric}$, so $\Box A - RA = 0$ type equations carry a Ricci mass term [20, 21]. Forman's discrete Bochner method [17] writes $L_1 = B_1 + F_1$ with F_1 diagonal edge curvature; for an unweighted 2-complex $F(e) = \#\text{faces}(e) + \#\text{vertices}(e) - \#\text{parallel neighbours}(e)$. On the square torus each edge has $2 + 2 - 4 = 0$: flat, no correction with unit or per-level stars. With data-derived stars $F \neq 0$ and $\omega_k^2 \to \omega_k^2 + \langle v_k, F_1 v_k\rangle_M$ is the curvature correction; Bochner's vanishing (if $F > 0$ everywhere, $\beta_1 = 0$) also explains regime (c). **Two polarizations.** The TT-gauge decomposition of a rank-2 wave into $h_+, h_\times$ [20] needs a symmetric, traceless, transverse tensor; a 2560×6912 weight matrix

has none of these, so no derivable structure follows. The nearest analogue is the singular value decomposition; it is not adopted.

4 DTFT formulation and estimators

For contributor i, band mode $m = (a, k)$, take the coefficient sequence along the opposite axis in MRH pop order, $s_{i,m}[n]$, and

$$S_{i,m}(\nu) = \sum_n w[n] s_{i,m}[n] e^{-j2\pi\nu n}, \quad \hat{\nu}_{i,m} = \arg\max_\nu |S_{i,m}|, \quad \phi_{i,m} = \arg S_{i,m}(\hat{\nu}_{\text{base},m}), \tag{D1}$$

phase taken at the base's frequency so models are compared at one off-grid point; group delay $-\partial_\nu \arg S$ is a sub-channel refinement of the phase-correlation shift (paper_v2 Eq. 12). **Coarse peak:** FFT [23] with a Hann window, gated by PSR ≥ 10 [31] and uniqueness ≥ 1.4, the gate already validated in the author's phase-correlation engine design (12-bit block floating point indistinguishable from float64 there; CORDIC phase-only arguments). Hann's -31.5 dB first sidelobe [24] is below the -20 dB (1/10) the PSR gate needs, with the narrowest mainlobe that does so. **Sub-bin:** rectangular three-sample estimator [25, 26]

$$\hat{\delta} = \frac{\tan(\pi/N)}{\pi/N} \Re \frac{X_{k-1} - X_{k+1}}{2X_k - X_{k-1} - X_{k+1}}, \qquad \hat{\nu} = (k + \hat{\delta})/N \tag{E1}$$

(sanity: true bin 7.3, estimate 7.30002, $N = 64$); Quinn [27] equivalent, Rife–Boorstyn [28] the ML/CRB reference. Off-grid single points by Goertzel [29] ($O(N)$), zooms by chirp-z [30].

5 Lock theory with derived thresholds

Ticks. A window is $N_{\text{lost}} = 255$ ticks (R-KTE-2); R-KTE-7 fixes one turn per f_K ticks for a free-running node. Mapping one pass over the sequence to one window,

$$\omega_{i,m} = \frac{(\hat{\nu}_{i,m} - \hat{\nu}_{\text{base},m})N}{N_{\text{lost}}} \text{ turn/tick}, \qquad \theta_{i,m}(0) = \phi_{i,m} - \phi_{\text{base},m}. \tag{L1}$$

(With N2 the same offset in eigenfrequency units is $\omega(\hat{\nu}_i) - \omega(\hat{\nu}_{\text{base}})$; the two agree to first order at small ν.) **Pair threshold (exact, linear lift).** Lag $e \leftarrow e + \Delta\omega - 2Ke$, steady $e^* = \Delta\omega/2K$; R-KTE-2 requires $|e^*| \leq W_{\text{lock}}$:

$$|\Delta\omega| \leq 2KW_{\text{lock}} = \tfrac{1}{128} \text{ turn/tick} \iff |\Delta\nu| N \leq N_{\text{lost}}/128 \approx 1.99. \tag{L2}$$

Sanity (tick simulation): $\Delta\omega = 1/256$ lag 0.031 locked; 1/128 lag 0.0625 (boundary) locked; 1/100 lag 0.08 not locked. **Population.** $K_c = 2/(\pi g(0))$ [3, 2]; Lorentzian half-width Δ: $K_c = 2\Delta$ and $r = \sqrt{1 - K_c/K}$ exactly (Ott–Antonsen [4]), oscillators with $|\omega| \leq Kr$ locked, others drift. $\Delta = \text{IQR}/2$ (Lorentzian quartiles). Now r_m in paper_v2 Eq. (19) is informative. **Merge decision.** Mode m of i enters S_m iff (i) PSR/uniqueness gate passes, (ii) (L2) at steady state, (iii) equal winding (H4), (iv) R-KTE-2 lost stays clear for N_{lost} ticks.

Algorithm D (selective lock, per target tensor; replaces steps 3–4 of paper_v2 Alg. 1).

1. Seed as paper_v2. Per-level stars (Sec. 2.4), λ_{gap} and band (H2), c_P/c_S (Sec. 3).
2. Persistence filtration by heap key; drop classes < 1 LSB; order stages by lifetime.
3. Per i, m: Hann FFT peak, PSR gate, (E1), Goertzel phase at $\hat{\nu}_{\text{base}}$; periods (H3); normal-mode check (N1)/(N2).
4. $\omega, \theta(0)$ by (L1); Δ, report K_c/K.
5. Tick R-KTE-1 *with* $s_i\omega_i$ and the $g_\omega = 8$ frequency update, stage by stage; record r_m, locked set, windings.
6. Merge by Eq. (19) over S_m; unlocked modes stay with base or route to adapters; emit $\hat{\nu}, \phi, q, p$ in the BNZ4 chunk.

6 Proposed changes to paper_v2 (not applied)

- Eq. (14b) $\omega_i = 0 \to$ (L1) with (N1)/(N2) as definition; Eq. (15) add the $s_i\omega_i$ term and ω update of R-KTE-1.
- Eq. (5) $\to$ (H2); Sec. 1: 2-cell note (Prop. 2); Eq. (4) star-weighted with per-level stars and c_P, c_S.
- Eq. (12) optionally refined by group delay; Eq. (16) $\to$ decision (i)–(iv); Eq. (18) $\to$ persistence order.
- Sec. 6 BNZ4: per mode $\hat{\nu}$ (BF16), ϕ (12-bit), winding (int8), periods (BF16). App. A: $\omega = 0$ row $\to$ Table here.

7 Complexity: 2560×6912, $C \approx 23$

Grid $G \le 256^2$ (paper_v2 Eq. 6), sequence $N \le 256$, band size B. Grid FFT per contributor $\approx G \log_2 G \approx 10^6$ ops; per mode one length-N FFT (≤ 2048) plus Goertzel (N). Front end $\approx 23(10^6 + 2.3 \times 10^3 B)$. Periods $O(C\beta_1 N)$. Normal modes: diagonal with per-level stars (free); LOBPCG on B vectors otherwise, $O(BG \log G)$ per iteration. Ticks $\approx 23\, B \cdot 255\, (D + 2)$ with bucket mean-field coupling. Persistence H_1 cost at $G = 256^2$ is a tuning measurement. These are operation counts, not timings.

8 Falsifiable predictions

P1 Identical source: $\Delta\nu = 0$, equal windings $\Rightarrow r = 1$, all locked.

P2 Randomly permuted source (or ternary noise at matched absmean): white sequences, most modes fail PSR ≥ 10; survivors have $\hat{\nu}$ near uniform over $N/2$ bins, so $P[(L2)] \approx 8/N$ (0.3% at $N = 2560$, 3% at 256); periods uncorrelated; r_m near the $1/\sqrt{n}$ floor.

P3 Digit-cyclic shift of base: periods unchanged, group delay recovers the shift, full lock after seeding.

P4 $M_\alpha = (1-\alpha)M_{\text{base}} + \alpha M_{\text{rand}}$: locked fraction falls monotonically, crossing where $2\Delta(\alpha) = K$.

P5 A 2B4T fine-tune locks on more modes than Qwen. If Qwen locks as often as the identical model, the method is falsified.

P6 Measured $\omega(\hat{\nu}_i)$ (N2) and Rayleigh-perturbed ω_k (N1) agree within estimator variance.

9 Tiny-scale validation protocol

Tiny files only; no numbers here are measured. Run P1–P4, P6 on 64×64 and 60×90 tensors: β_1 by exact rank vs Prop. 2; estimator error vs N against the Rife–Boorstyn CRB; locked fraction, r, K_c/K, windings, Hodge split; 10 permutation seeds for P2 with the empirical 95th percentile as acceptance threshold. Then P5 on one 2B4T layer vs Qwen. Arithmetic checks run so far are in `dtft_lock_sanity.py`.

A Parameter derivations

Parameter	Value	Derivation / source
natural frequency	(N1)/(N2)	normal modes, $K = \star_1 L_1^\star$, $M = \star_1$ (author's framework; Goldstein)
measured $\omega_{i,m}$	(L1)	DTFT offset; R-KTE-7 scaling, $N_{\text{lost}} = 255$ (R-KTE-2)
$\theta(0)$	$\phi_i - \phi_{\text{base}}$	DTFT argument at base frequency
K phase/freq	$2^{-4}/2^{-8}$	KTE g_θ, g_ω (R-KTE-1)
pair bound	1/128 turn/tick	(L2) $= 2KW_{\text{lock}}$, $W_{\text{lock}} = 1/16$ turn
K_c; Δ	$2/(\pi g(0))$; IQR/2	Kuramoto; Ott–Antonsen Lorentzian
band edge	(H2)	first non-zero eigenvalue, coarsest long cycle
stars	per-level energy	DFT-commutation constraint; L^2 metric (AFW)
c_P/c_S	$\sqrt{R_g/R_c}$	Lamé split (Landau–Lifshitz); base Rayleigh quotients
curvature	F_1 (Forman)	zero on square torus; nonzero only with data stars
window	Hann	least mainlobe meeting PSR -20 dB (Harris)
PSR, uniqueness	10, 1.4	author's phase-correlation engine design
word length	12-bit BFP	same design
sub-bin	(E1)	Jacobsen–Kootsookos; Candan
persistence floor	1 key LSB	12-bit MRH key (paper_v2 Eq. 7)

Tuning items: sequence axis when both axes carry long digits (alternative: 2-D DTFT); non-Lorentzian g (K_c via kernel density, bandwidth is a choice); inhomogeneous activation-based stars (calibration data, LOBPCG); multi-peak modes (one oscillator per peak?); H_1 persistence cost at 256^2; twisted-state basins when windings differ by one; whether c_P/c_S should be per layer or per role.

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